

Лабораторная работа № 15

Настройка сетевого журналирования

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Цель:

Получить навыки централизованного сбора и просмотра системных журналов.



Приём rsyslog по TCP

```
Выбрать root@server:/etc/rsyslog.d
rpcbind 11528 rpc 5u IPv4 34106 0t0 TCP *:
sunrpc (LISTEN)
rpcbind 11528 rpc 7u IPv6 34120 0t0 TCP *:
sunrpc (LISTEN)
rpc.statd 11534 rpcuser 8u IPv4 34219 0t0 TCP *:
34621 (LISTEN)
rpc.statd 11534 rpcuser 10u IPv6 34225 0t0 TCP *:
45419 (LISTEN)
rpc.mount 12845 root 5u IPv4 80942 0t0 TCP *:
mountd (LISTEN)
rpc.mount 12845 root 7u IPv6 80946 0t0 TCP *:
mountd (LISTEN)
sshd-session 12926 root 4u IPv4 81696 0t0 TCP se
rver:ssh->_gateway:59467 (ESTABLISHED)
sshd-session 12930 vagrant 4u IPv4 81696 0t0 TCP se
rver:ssh->_gateway:59467 (ESTABLISHED)
smbd 14163 root 27u IPv6 93950 0t0 TCP *:
microsoft-ds (LISTEN)
smbd 14163 root 28u IPv6 93951 0t0 TCP *:
netbios-ssn (LISTEN)
smbd 14163 root 29u IPv4 93952 0t0 TCP *:
microsoft-ds (LISTEN)
smbd 14163 root 30u IPv4 93953 0t0 TCP *:
netbios-ssn (LISTEN)
smbd[192. 14174 root 32u IPv4 92987 0t0 TCP ww
w.ndbaranov.net:microsoft-ds->client.ndbaranov.net:33856 (ESTABLISHED)
rsyslogd 14258 root 4u IPv4 95690 0t0 TCP *:
shell (LISTEN)
rsyslogd 14258 root 5u IPv6 95691 0t0 TCP *:
shell (LISTEN)
rsyslogd 14258 14260 in:imjour root 4u IPv4 95690 0t0 TCP *:
shell (LISTEN)
rsyslogd 14258 14260 in:imjour root 5u IPv6 95691 0t0 TCP *:
shell (LISTEN)
rsyslogd 14258 14261 in:imtcp root 4u IPv4 95690 0t0 TCP *:
shell (LISTEN)
rsyslogd 14258 14261 in:imtcp root 5u IPv6 95691 0t0 TCP *:
shell (LISTEN)
rsyslogd 14258 14262 w0/imtcp root 4u IPv4 95690 0t0 TCP *:
shell (LISTEN)
rsyslogd 14258 14262 w0/imtcp root 5u IPv6 95691 0t0 TCP *:
shell (LISTEN)
rsyslogd 14258 14263 w1/imtcp root 4u IPv4 95690 0t0 TCP *:
shell (LISTEN)
rsyslogd 14258 14263 w1/imtcp root 5u IPv6 95691 0t0 TCP *:
shell (LISTEN)
rsyslogd 14258 14264 rs:main root 4u IPv4 95690 0t0 TCP *:
shell (LISTEN)
rsyslogd 14258 14264 rs:main root 5u IPv6 95691 0t0 TCP *:
shell (LISTEN)
root@server:/etc/rsyslog.d# lsof -iTCP:514 -sTCP:LISTEN
COMMAND PID USER FD TYPE DEVICE SIZE/OFF NODE NAME
rsyslogd 14258 root 4u IPv4 95690 0t0 TCP *:shell (LISTEN)
rsyslogd 14258 root 5u IPv6 95691 0t0 TCP *:shell (LISTEN)
root@server:/etc/rsyslog.d#
```

Смысл результата

На server загружен модуль imtcp и включён приём сообщений на порту 514. Проверка сокета показывает, что rsyslog действительно слушает TCP.

Правило firewalld для syslog

```
root@server:/etc/rsyslog.d# firewall-cmd --add-port=514/tcp
success
root@server:/etc/rsyslog.d# firewall-cmd --add-port=514/tcp --permanent
success
root@server:/etc/rsyslog.d# firewall-cmd --list-ports
514/tcp 2022/tcp
root@server:/etc/rsyslog.d#
```

Смысл результата

TCP-порт 514 добавлен в постоянную конфигурацию firewalld. После reload правило присутствует в списке открытых портов.

Пересылка журналов с client

```
LOG
2026-09-06T22:24:36 UTC Press F2 to enable mouse support
LOG 2026-09-06T22:24:34.000 : syslog_log : messages[7,661] : systemd[1] :
Sep 06 22:22:49 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter
Sep 06 22:22:49 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.
Sep 06 22:22:59 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.
Sep 06 22:22:59 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter
Sep 06 22:22:59 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.
Sep 06 22:23:09 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.
Sep 06 22:23:09 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter
Sep 06 22:23:09 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.
Sep 06 22:23:19 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.
Sep 06 22:23:19 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter
Sep 06 22:23:19 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.
Sep 06 22:23:26 server systemd[1]: Starting packagekit.service - PackageKit Daemon...
Sep 06 22:23:26 server systemd[1]: Started packagekit.service - PackageKit Daemon.
Sep 06 22:23:29 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.
Sep 06 22:23:30 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter
Sep 06 22:23:30 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.
Sep 06 22:23:40 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.
Sep 06 22:23:40 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter
Sep 06 22:23:40 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.
Sep 06 22:23:50 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.
Sep 06 22:23:50 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter
Sep 06 22:23:50 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.
Sep 06 22:24:00 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.
Sep 06 22:24:00 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter
Sep 06 22:24:00 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.
Sep 06 22:24:03 server systemd[1]: Starting systemd-hostnamed.service - Hostname Service...
Sep 06 22:24:04 server systemd[1]: Started systemd-hostnamed.service - Hostname Service.
Sep 06 22:24:10 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.
Sep 06 22:24:11 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter
Sep 06 22:24:11 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.
Sep 06 22:24:17 server systemd[1]: Started run-p14545-i14546.service - [systemd-run] /usr/bin/systemd
Sep 06 22:24:17 server systemd[1]: Starting man-db-cache-update.service...
Sep 06 22:24:21 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.
Sep 06 22:24:21 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter
Sep 06 22:24:21 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.
Sep 06 22:24:21 server systemd[1]: man-db-cache-update.service: Deactivated successfully.
Sep 06 22:24:21 server systemd[1]: Finished man-db-cache-update.service.
Sep 06 22:24:21 server systemd[1]: man-db-cache-update.service: Consumed 2.438s CPU time, 28.4M memor
Sep 06 22:24:21 server systemd[1]: run-p14545-i14546.service: Deactivated successfully.
Sep 06 22:24:31 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.
Sep 06 22:24:32 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter
Sep 06 22:24:32 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.
Sep 06 22:24:34 server systemd[1]: systemd-hostnamed.service: Deactivated successfully.
```

Смысл результата

В клиентском конфигурационном файле задана отправка *.* на @@server.ndbaranov.net:514. Двойной символ @ означает надёжную передачу по TCP.

Сообщения клиента на сервере

```
Администратор: Windows PowerShell
Windows PowerShell
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PS C:\Windows\system32> notepad .\provision\server\netlog.sh
>> cd D:\work\ndbaranov\vagrant
PS D:\work\ndbaranov\vagrant> cd D:\work\ndbaranov\vagrant
PS D:\work\ndbaranov\vagrant> notepad .\provision\server\netlog.sh
PS D:\work\ndbaranov\vagrant> notepad .\provision\client\netlog.sh
PS D:\work\ndbaranov\vagrant> vagrant validate
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
Vagrantfile validated successfully.
PS D:\work\ndbaranov\vagrant> vagrant provision server --provision-with "server netlog"
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Running provisioner: server netlog (shell)...
server: Running: C:/Users/barash/AppData/Local/Temp/vagrant-shell120260907-25516-jj1vc5.sh
server: Provisioning network log server
server: Install needed packages
server: Last metadata expiration check: 0:33:48 ago on Sun 06 Sep 2026 09:53:03 PM UTC.
server: Package rsyslog-8.2510.0-5.el10.x86_64 is already installed.
server: Dependencies resolved.
server: Nothing to do.
server: Complete!
server: Configure rsyslog
server: Configure firewall
server: Warning: ALREADY_ENABLED: '514:tcp' already in 'public'
server: success
server: success
server: Warning: ALREADY_ENABLED: 514:tcp
server: Restart rsyslog service
server: Network log server provisioning completed
PS D:\work\ndbaranov\vagrant> vagrant provision client --provision-with "client netlog"
==> client: Running provisioner: client netlog (shell)...
client: Running: C:/Users/barash/AppData/Local/Temp/vagrant-shell120260907-22336-3rc4eb.sh
client: Provisioning network log client
client: Install needed packages
client: Last metadata expiration check: 0:18:13 ago on Sun 06 Sep 2026 10:09:22 PM UTC.
client: Package rsyslog-8.2510.0-5.el10.x86_64 is already installed.
client: Dependencies resolved.
client: Nothing to do.
client: Complete!
client: Last metadata expiration check: 0:18:14 ago on Sun 06 Sep 2026 10:09:22 PM UTC.
client: Dependencies resolved.
client: =====
client: Package      Architecture Version      Repository Size
client: =====
client: Installing:
client: lnav            x86_64      0.13.2-1.el10_2 epel        3.6 M
client:
client: Transaction Summary
```

Смысл результата

После перезапуска rsyslog записи с hostname клиента появились в центральном журнале server. Provision-сценарии закрепляют настройки обеих сторон.

Результат

rsyslog принимает сообщения клиента по TCP/514 и сохраняет их централизованно. Наличие записей с hostname client подтверждает работу пересылки.